Giulia Ischia

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Contact Information

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Research Experience

2026 – now **Postdoctoral Researcher, CNRS-Ircelyon**, Lyon, France
Supervisor: Dr. D. Laurenti.
Independent research project funded by the MOPGA Postdoctoral Fellowship (€30'500). Topic: sustainable conversion of lignin into advanced nanocarbons for catalytic applications.

2024 – 2026 **Postdoctoral Researcher, Max Planck Institute of Colloids and Interfaces**, Potsdam, Germany
Supervisor: Prof. Dr. M. Antonietti
Independent research project funded by the Alexander von Humboldt Foundation (€90'000) on the hydrothermal conversion of biomass into artificial humic substances for environmental applications.

2022 – 2023 **Postdoctoral Researcher**, University of Trento, Trento, Italy
Supervisor: Prof. L. Fiori.
Research on hydrothermal conversion of lignin and bioplastic waste into bio-based products.

Education

2022 **PhD in Engineering (major: Chemistry and Materials)**
University of Trento, Trento, Italy
- Focus on: hydrothermal conversion of biomass, reaction mechanisms, biofuels, kinetic modelling, realization of prototype systems integrating hydrothermal and solar systems.
- Thesis title: "Sustainable conversion of biomass wastes via hydrothermal processes: fundamentals and technology" (*supervisors: Prof. L. Fiori and Prof. A. Miotello*).

2018 **MSc in Energy Engineering**
University of Trento and Free University of Bolzano, Trento/Bolzano, Italy
- Specialization: renewable energy and biomass. Grade: 110/110 cum laude.
- Thesis title: "Sustainable conversion of biomass wastes via hydrothermal processes: fundamentals and technology" (*supervisors: Prof. L. Fiori and Prof. J.L. Goldfarb*).

2015 **Bachelor in Energy Engineering**
Politecnico di Torino, Torino, Italy

Visiting Researcher

- **Northeast Agricultural University**, Harbin, China (*host: Prof. F. Yang*) **May 2025 – June 2025**
- **Tohoku University**, Sendai, Japan (*host: Prof. T. Adschiri*) - Tohoku-AIMR fellowship **Aug. 2023 – Nov. 2023**
- **Cornell University**, Ithaca, USA (*host: Prof. J.L. Goldfarb*) - Fulbright fellowship **March 2021 – Nov. 2021**
- **Boston University**, Boston, USA (*host: Prof. J.L. Goldfarb*) **Sept. 2017 – Jan. 2018**

Academic Teaching

- **Visiting Assistant Professor, Northeast Agricultural University**, Harbin, China **2025**
- **Teaching Assistant at University of Trento**, Trento, Italy **2019 – 2023**

Research fundings and awards

Research fundings

- **MOPGA (Make Our Planet Great Again) Postdoctoral Fellowship**, by Campus France (€30'500). Role: PI. CNRS-IRCELYON, Lyon, France. [2026–2027]
- **Humboldt Research Fellowship for Postdoctoral Researchers**, by the Alexander von Humboldt Foundation (€90'000). Role: PI. Max Planck Institute of Colloids and Interfaces, Germany. [2024–2026]
- **International Researcher Visiting Fellowship**, by Tohoku University - Advanced Institute for Materials Research (WPI-AIMR) ($5'000). Role: PI. Tohoku University, Sendai, Japan. [2023]
- **Fulbright Research Scholar Fellowship**, by the U.S. Fulbright Commission ($11'000). Role: PI. Cornell University, USA. [2021]

Awards

- **ACS Sustainability Star**, by the American Chemical Society (ACS) (Top 30 globally). [2024]
- **Best Young Researcher Award**, at the WasteEng Conference (€300). [2021]
- **Joseph Breen Memorial Fellowship**, by the American Chemical Society (ACS) ($2'000). [2021]
- **The Journey Fellowship**, by EIT Climate-KIC (€5'000). [2020]

Other research-related activities

- **Reviewer** for > 20 journals, including Chemical Engineering Journal, Food Chemistry, Waste Management, Nature Communications, Bioresource Technology, Fuel, Journal of Environmental Chemical Engineering, Journal of Applied & Analytical Pyrolysis, ACS Sustainable Chemistry & Engineering.
- **Outreach and Communication:** participated in "Women in Science" initiative for secondary schools in Northern Italy (2024) and European Researchers' Night in Trento (2022-2024); Developed research project websites.

Skills

- **Laboratory and analytical equipment**: advanced user of chromatography (GC-MS, Py-GC-MS, HPLC), mass spectrometry (ICP-MS), NMR (liquid and solid-state), spectroscopy (FTIR, Raman, UV–Vis), microscopy (optical, SEM, TEM), thermal analysis (TGA, DSC), elemental analysis (CHNS), TOC, BET, and XRD. Hands-on experience with high-pressure and hydrothermal reactors, laboratory furnaces.
- **Software**: advanced proficiency in Python, MATLAB, Aspen Plus, OriginPro, MS Office, LaTeX, SolidWorks, 3D CAD modelling, ChemDraw.
- **R&D Project Coordination**: Interdisciplinary project management, grant and proposal writing, technical reporting, mentoring and supervision of students and researchers, collaboration with academic and industrial partners.
- **Languages**: Italian (native), English (C1), German (B1), French (A2).

Presentations at international conferences

- **International Symposium on Green Chemistry – ISGC**. *LaRochelle, France*. Oral presentation: "From biomass to artificial humic substances: a new path for soil health and carbon sequestration". May 2025.
- **8th Expert Forum on Hydrothermal Processes**. *Leipzig, Germany*. Oral presentation: "From hydrothermal carbonization to humification for soil carbon sequestration". November 2025.
- **Green Chemistry (GRS) Gordon Research Seminar**. *Castelldefels, Spain*. Oral presentation: "From biomass to artificial humic matter for soil carbon sequestration". July 2024. ***Invited speaker***.
- **International Solvothermal and Hydrothermal Synthesis Association seminar series**. Virtual. June. Oral presentation: "From biomass to artificial humic matter for soil carbon sequestration". April 2024. ***Invited Speaker.***
- **27th Annual Green Chemistry & Engineering Conference.** *Long Beach, USA*. June. Oral presentation: "Hydrothermal carbonization as green route for valorizing bioplastic wastes". June 2023. ***Invited Speaker***.

- **3rd Symposium on Hydrothermal Carbonization**. *Seoul, South Korea*. Oral presentation: "Kinetics and chemistry behind the transition from hydrothermal carbonization to liquefaction". May 2023
- **American Chemical Society (ACS) Spring**. *Virtual*. February. Oral presentation: "From hydrothermal carbonization to liquefaction: The chemistry and kinetics along the hydrothermal biomass reaction spectrum". February 2022.
- **8th International Conference on Engineering for Waste and Biomass Valorisation**. *Virtual*. Oral presentation: "Sun to biocoal: a solar - hydrothermal system for waste biomass valorization". June 2021. ***Best young researcher award***.
- **24th Annual Green Chemistry & Engineering Conference - Systems Inspired Design**. *Virtual*. Oral presentation: "Post-treatments after hydrothermal carbonization: From organic wastes to solid and liquid biofuels". June 2020.
- **2nd International Symposium on Hydrothermal Carbonization**, *Berlin, Germany*. Poster presentation: "Realization of a solar hydrothermal reactor: A hybrid solution to develop a zero-energy technology". May 2019.

Peer-reviewed publications

(* indicates corresponding author)

[24] Pakzad, K., **Ischia, G.**, Antonietti, M., Kim, J., Rotter, V.S., Volikov, A. (2026). *Cation Modification of Montmorillonite Regulates the Retention of Biomass-Derived Artificial Humic Substances*. RSC Advances. In press. [10.1039/d6ra09407d]

[23] Akogun, O., **Ischia, G.**, Mohammadebrahimi, M., Scheufele, F. B., Román, S., Berge, N. D., Fiori, L., Libra, J. A., Rotter, V. S., Hoffmann, T., Marzban, N. (2026). *Controlled transition from hydrothermal carbonization to fulvification for artificial humic substance production from wheat straw*. Chemical Engineering Journal, 182557. In press. [10.1016/j.cej.2026.182557]

[22] Ghobadian S., Schneider R., **Ischia G.**, Fischer M., Romero O.,Venus, J., Hoffmann, T., Sturm, B., Rotter S., Kraume M., Marzban, M. (2026). *Toward Waste-Minimising Biorefineries: Integrating Fermentation and Hydrothermal Fulvification of Grass*. Waste Management. 221.115625. [10.1016/j.wasman.2026.115625]

[21] Sarlaki, E., Nia A., Hu, S, Marzban, M., **Ischia, G.**, Jia, M., Rezaei, M., Ahmadi, Javaheri, P., Malekian, A., Tabatabaei, M., Aghbashlo, M. (2026). *Progress and challenges in thermochemical technologies for biomass humification: A comprehensive review*. Renewable and Sustainable Energy Reviews, 232, 116775. [10.1016/j.rser.2026.116775]

[20] Ghobadian, S., **Ischia, G.**, Romero, O., Rossi, G., Hoffmann, T., Kraume, M., Marzban, N. (2026). *Hydrothermal Humification and Fulvification of Grass for Artificial Humic Substance Production and By-product Applications*. Environmental Technology & Innovation, 41, 104836. [10.1016/j.eti.2026.104836]

[19] **Ischia, G.***, Marzban, N., Schmidt, J., Volikov, A. (2026). *Transitioning from hydrothermal carbonization to humification for producing artificial humic substances*. Bioresource Technology, 439, 13336. [10.1016/j.biortech.2025.133306]

[18] **Ischia, G.***, Yoko, A., Wahyudiono, Fiori, L., Adschiri, T. (2025). *Valorization of lignin into carbon spheres, carbon dots, and graphene-like layers via hydrothermal conversion*. International Journal of Biological Macromolecules, 330, 147997. [10.1016/j.ijbiomac.2025.147997]

[17] Zhang, Z., Xie, D., Cheng, K., Liu Z., **Ischia, G.**, Zhao, Y., Yang, F. (2025). *Optimization of Fertilization Strategies for Paddy Fields based on Multi-task Transfer Learning and Reinforcement Learning*. Agricultural Water Management, 322, 109965. [10.1016/j.agwat.2025.109965]

[16] Queffelec, J., Torres Pérez, M.D., Domínguez, H., **Ischia, G.**, Filonenko, S. (2025). *Tailoring the extraction process and properties of polysaccharides from the lichen Evernia prunastri using natural eutectic solvents following a biorefinery approach*. Green Chemistry, 27, 6493–6511. [10.1039/d5gc01413a]

[15] **Ischia, G.**, Marchelli, F., Bazzanella, N., Ceccato, R., Calvi M., Guella, G., Gioia, C., Fiori, L. (2024). *Cellulose Acetates in Hydrothermal Carbonization: a Green Pathway to Valorize Residual Bioplastics*. ChemSusChem, 18, e202401163. [10.1002/cssc.202401163]

[14] **Ischia, G.**, Sudibyo, H., Miotello, A., Tester J.W., Fiori, L., Goldfarb, J.L. (2024). *Identifying the Transition from Hydrothermal Carbonization to Liquefaction of Biomass in a Batch System*. ACS Sustainable Chemistry & Engineering, 12, 4539-4550. [10.1021/acssuschemeng.3c07731]

[13] **Ischia, G.**, Berge, N.D., Bae, S., Marzban, N., Román, S., Farru, G., Wilk, M., Kulli, B., Fiori, L. (2024). *Advances in Research and Technology of Hydrothermal Carbonization: Achievements and Future Directions*. Agronomy 14(5), 955. [10.3390/agronomy14050955]

[12] **Ischia, G.**, Goldfarb, J.L., Miotello, A., Fiori, L. (2023). *Green solvents to enhance hydrochar quality and clarify effects of secondary char*. Bioresource Technology, 338, 129724. [10.1016/j.biortech.2023.129724]

[11] **Ischia, G**., Marchelli, F., Fiori, L. (2023). Advancing bioenergy for a green future. Detritus, 23, 1–2. [10.31025/2611-4135/2023.18278]

[10] Marchelli, F., Ferrentino, R., **Ischia, G.**, Calvi, M., Andreottola, G., Fiori, L. (2023). *Valorisation of eyewear bioplasics through HTC and anaerobic digestion: preliminary results*. Detritus, 23, 35–42. [10.31025/2611-4135/2023.18275]

[9] Marchelli, F., **Ischia, G.**, Calvi, M., Fiori, L. (2023). *Hydrothermal Carbonization as a Process to Facilitate the Disposal of Bioplastics*. Chemical Engineering Transactions, 99, 79–84. [10.3303/CET2399014]

[8] Battocchio, P., Bazzanella, N., Orlandi, M., **Ischia, G**., Miotello, A. (2023). *Carbon nanoparticles as absorbers in PVC for laser ablation propulsion: size effects*. Applied Physics A: Materials Science and Processing, 129, 1–12. [10.1007/s00339-023-06502-7]

[7] **Ischia, G.**, Cutillo, M., Guella, G., Bazzanella, N., Cazzanelli, M., Orlandi, M., Miotello, A., Fiori, L. (2022). *Hydrothermal carbonization of glucose: Secondary char properties, reaction pathways, and kinetics*. Chemical Engineering Journal, 449, 137827. [10.1016/j.cej.2022.137827]

[6] **Ischia, G.***, Cazzanelli, M., Fiori, L., Orlandi, M., Miotello, A. (2022): *Exothermicity of hydrothermal carbonization: Determination of heat profile and enthalpy of reaction via high-pressure differential scanning calorimetry*. Fuel, 310, 122312. [10.1016/j.fuel.2021.122312]

[5] Gonnella, G., **Ischia, G.**, Fambri, L., Fiori, L. (2022). *Thermal Analysis and Kinetic Modeling of Pyrolysis and Oxidation of Hydrochars*. Energies, 15, 1–21. [10.3390/en15030950]

[4] **Ischia, G**., Fiori, L. (2021): *Hydrothermal Carbonization of Organic Waste and Biomass. A Review on Process, Reactor, and Plant Modeling*. Waste and Biomass Valorization, 12, 2797–2824. [10.1007/s12649-020-01255-3]. **Most cited article of the journal in 2022.**

[3] **Ischia, G.**, Fiori, L., Gao, L., Goldfarb, J.L. (2021). *Valorizing municipal solid waste via integrating hydrothermal carbonization and downstream extraction for biofuel production*. Journal of Cleaner Production, 289, 125781. [10.1016/j.jclepro.2021.125781]

[2] **Ischia, G.**, Castello, D., Orlandi, M., Miotello, A., Rosendahl, L.A., Fiori, L. (2020). *Waste to biofuels through zero-energy hydrothermal solar plants: Process design*. Chemical Engineering Transactions, 80, 7–12. [10.3303/CET2080002]

[1] **Ischia, G.**, Orlandi, M., Fendrich, M.A., Bettonte, M., Merzari, F., Miotello, A., Fiori, L. (2020). *Realization of a solar hydrothermal carbonization reactor: A zero-energy technology for waste biomass valorization*. Journal of Environmental Management, 259, 110067. [10.1016/j.jenvman.2020.110067]